Project Classes

FIRST MEETING · RESEARCH, RESPONSIBILITY AND REPRODUCIBLE EVIDENCE

From curiosity to credible evidence

One or two students. One research question.
A project worth understanding — and perhaps publishing.

Adrian Horzyk
horzyk@agh.edu.pl

OUR RESEARCH CONTRACT

01 Question & literature

02 Experiment & evidence

03 Understanding & ownership

Teams code:
ljbb9ii

Home message: Ask a focused question. Build a fair experiment. Explain what the evidence means.

The goal: a question answered with evidence

SEVERAL RELEVANT APPROACHES · FAIR COMPARISONS · A JUSTIFIED CONCLUSION

Your goal is not to build as many models as possible but to answer an interesting computational intelligence question through a well-designed and smart computational experiment, supported by reproducible evidence and fair comparisons of several relevant state-of-the-art approaches, models or algorithms.

A STRONG PROJECT SHOULD CONTAIN

01 Clear research question → **02 Evidence from literature** → **03 Reproducible baseline** → **04 Meaningful modification or comparison**

05 Rigorous evaluation

06 Interpretation of successes and failures

07 Conclusions supported by evidence

Home message: Model count is secondary; comparisons must be strong enough to support the claim.

Turn a topic into a research question

DEFINE THE VARIABLE, THE COMPARISON AND THE EVIDENCE THAT WOULD CHANGE YOUR MIND

Vision Transformers	Which is more robust with 10% labels: a CNN, a ViT or frozen pretrained features? Control: identical splits, label subsets and tuning budget.
LLM self-reflection	Does critique or verification improve answers when every method gets the same token budget? Control: fixed questions, verified answers and compute reporting.
Associative memory	How does associative retrieval compare with attention under noise, partial cues and new insertions? Control: identical stored items, queries and memory budget.

Reproduce · Extend · Stress-test · Transfer · Hybridise · Compress · Explain

Include a meaningful baseline and sufficient comparisons; justify the design at Class 3.

Home message: Choose an anchor paper and a research action; then narrow the scope.

Research challenges I: models & adaptation

LECTURES 2–5 · SMALL MODELS CAN ANSWER IMPORTANT QUESTIONS

L2 How much supervision is enough?

Compare supervised, self-supervised and transfer learning as labels decrease.

L3 Adaptation under a compute budget

Compare frozen features, adapters/LoRA and full fine-tuning on a small model.

L3 + L8 Compression without losing trust

Test whether quantisation or distillation changes calibration and robustness.

L4 CNN vs ViT in difficult conditions

Compare architectures under small data, corruption or domain shift.

L5 When do graph models help?

Compare GNNs with strong non-graph baselines and remove graph information.

L5 + L13 When does synthetic data help?

Vary synthetic/real ratios and evaluate a downstream model on real data.

Home message: Choose a phenomenon to study; use models as experimental instruments.

Research challenges II: agents, trust & action

LECTURES 6–9 · MEASURE RELIABILITY, GENERALISATION AND COST

L6 **World model vs reactive agent**

Test whether predicting the environment improves a small agent's planning.

L7 **Can curiosity help sparse rewards?**

Compare extrinsic-only and intrinsic-reward agents in a compact environment.

L8 **Knowing when the model does not know**

Compare calibrated confidence, MC dropout or small ensembles under shift.

L8 **Are explanations faithful and stable?**

Test SHAP, integrated gradients or counterfactuals under controlled changes.

L9 **Can discovered patterns improve prediction?**

Add mined patterns or rules to a strong baseline and ablate those features.

L6–L9 **Reliable compact multimodal systems**

Compare text-only, image-only and joint representations under missing input.

Home message: A fair stress test often teaches more than another leaderboard score.

Research challenges III: knowledge & frontier CI

LECTURES 10–14 · MEMORY, RULES, GENERATION AND REFLECTION

L10–L11 Associative memory vs attention

Compare associative retrieval, a Hopfield-style memory and attention.

L12 Learning with rules and constraints

Test whether domain rules improve data efficiency or constraint satisfaction.

L13 Generative models as scientific tools

Use a small generative model for anomalies or counterfactuals, then validate.

L14 Does self-reflection actually work?

Compare direct answers, critique and verification on a fixed question set.

L3, 10–11 Continual learning without forgetting

Compare fine-tuning, replay and associative memory across sequential tasks.

L12 + L14 Can a verifier improve reasoning?

Compare neural answers, rule checks and a hybrid verifier on small tasks.

Home message: Ambition comes from the question and evidence, not from hardware size.

One project. Individual reading and ownership.

TEAMS OF 1–2 · AT LEAST 3 DISTINCT RESEARCH PAPERS PER STUDENT BY CLASS 3

SOLO: 3 PAPERS

One focused research question.
A baseline and justified comparisons.
Substantive work you can explain.

PAIR: 6 DIFFERENT PAPERS

3 different papers assigned per student.
Both know the common anchor paper.
Two substantive, comparable work packages.

KEEP ONE SHORT LITERATURE MATRIX

Reader	Paper + link	Claim & evidence	Limitation	Project use

Summaries alone do not count. Prefer accepted research papers; verify status and citations.

Home message: A paper counts when you can explain its claim, evidence, limitation and use in your project.

Where to find strong open research: ML & AI

OFFICIAL PROCEEDINGS FIRST · CLICK THE COLOURED SOURCE NAMES

CORE MACHINE LEARNING

NeurIPS archive
Latest archived edition: 2025.

ICLR 2026 · OpenReview
Check acceptance and read the reviews.

ICML 2026 · PMLR v306
Open papers, supplements and linked code.

PMLR proceedings directory
AISTATS, UAI, COLT, CLeaR, CoRL and more.

AI, STRUCTURE & CONTROL

AAAI proceedings
Learning, planning, reasoning and agents.

IJCAI 2026
Broad AI; use relevant main-track sections.

CoRL · RSS · L4DC
Robot learning, embodiment and control.

Learning on Graphs · NeSy
Specialist graph and neuro-symbolic research.

Full guide: 29 venue entries, direct links and topic suggestions; use the latest released edition.

Home message: Find your own anchor paper; choose relevance, credible evaluation and feasibility.

More open sources: vision, language & journals

USE THE SOURCE THAT MATCHES THE RESEARCH QUESTION

VISION

CVPR · ICCV · WACV
ECCV / ECVA
Images, multimodal models, robustness and generation.

LANGUAGE

ACL · EMNLP · NAACL
TACL journal
Language models, efficient adaptation and evaluation.

BIOMEDICAL

MIDL 2026
MLHC · CHIL
Medical imaging, health data and clinical evaluation.

OPEN JOURNALS

JMLR · TMLR
JAIR · DMLR
Learning, general AI and data-centric research.

arXiv and discovery tools can help you search. Label preprints; follow links to the original paper.

Home message: Open access describes availability; assess the paper's methods and publication status.

Research hygiene: make the evidence trustworthy

A SMALL, FAIR EXPERIMENT IS MORE VALUABLE THAN A LARGE, UNCONTROLLED ONE

Separate data

Fixed train/validation/test splits. Fit preprocessing on training data only; prevent leakage.

Compare fairly

Same test protocol and fair tuning budgets; report pretraining, data and compute differences.

Test the claim

Task-relevant metrics, a baseline and a targeted ablation, stress test or equivalent control.

Quantify variation

For key stochastic comparisons, aim for ≥3 affordable seeds; label the spread, method and n.

Record the experiment

Save settings, seeds, raw metrics and versions. Report hardware, runtime and any API cost.

Interpret honestly

Analyse failures and limitations. A rigorous negative result can earn an excellent grade.

Home message: Negative results are also welcome when the experiment and interpretation are sound.

Use AI tools. Take responsibility for the work.

AI ASSISTANCE IS ALLOWED · UNDERSTANDING AND ACCOUNTABILITY REMAIN YOURS

Use AI productively

Brainstorming, explanations, debugging, coding support and language editing are allowed.

Keep intellectual ownership/contribution

Explain and modify your implementation; defend the choices that produced your results.

Verify every output

Read cited papers. Check references, code, formulas and claims against original evidence.

Preserve research integrity

Never invent results, references or contributions. Plot real outputs; label synthetic data.

Protect data and credit work

Respect licences. Keep private or confidential materials out of unapproved services.

Disclose briefly

3–5 lines: tool/model, purpose, affected parts and checks. No prompt diary is required.

Home message: We assess what you understand, verify and contribute — not whether you used an assistant.

Seven meetings: a compact research process

FORMATIVE CHECKPOINTS · ONE EVOLVING PRESENTATION · ONE FINAL RUBRIC

1	**Challenge & inspiration**	Understand rules; identify interests and feasible resources.
2	**Research pitch**	1–3 ideas; draft question; anchor paper; data; baseline; roles.
3	**Literature → protocol**	3 papers/person; hypothesis; splits; metrics; comparison plan.
4	**Baseline & data check**	Working baseline; data audit; first reproducible evaluation.
5	**First research results**	First hypothesis test; failures; next controlled experiment.
6	**Internal research review**	Ablation/stress test; uncertainty; feedback; draft final presentation.
7	**Final talk & submission**	Final presentations AND submission of assessed projects.

Final slots: 5–6 min/person, including brief questions; a pair receives 10–12 min.

Home message: Develop one presentation throughout the semester and submit its complete final version.

One project presentation throughout the semester

THE COMPLETE DECK DOCUMENTS THE RESEARCH; THE FINAL TALK USES SELECTED SLIDES

Question and motivation
Problem, research question, hypothesis and why the study matters.

Literature and studied methods
Explain the papers, concepts, algorithms and learning approaches you studied.

Data and experimental design
Datasets, preprocessing, splits, baselines, comparisons, metrics and resource limits.

Models and the final solution
Architectures, implementation, training, adaptation, optimisation and regularisation.

Results and visual evidence
Training/validation curves, comparison tables, illustrations and task-relevant diagnostics.

Interpretation of experiments
Explain successes, failures, ablations, uncertainty and differences between methods.

Conclusions and summary
Answer the question, state limitations and propose justified next steps.

References and ownership/contribution
Citations, code/result links, each person's contribution and a brief AI-use statement.

A contents slide and a concise overview make the full presentation easy to review.

Home message: Develop the same presentation from your first research pitch to the final solution.

A transparent rubric: 80 team + 20 individual

SAME CRITERIA FOR SOLO AND PAIR PROJECTS · SCORE EACH CRITERION ON A 0–4 SCALE

Criterion	Pts	Who	Evidence to inspect
Question & literature	**15**	Team	Slides: question, literature, rationale; reading matrix.
Design & comparisons	**20**	Team	Baseline, comparators, splits, metrics and fair budgets.
Implementation & reproducibility	**15**	Team	Working code, README, exact commands and saved runs.
Results & interpretation	**20**	Team	Full deck: experiments, plots, interpretation and limits.
Communication & submission	**10**	Team	Complete semester deck, clear visuals and submission.
Individual understanding	**10**	Each	Explain decisions, methods and the meaning of results.
Individual contribution	**10**	Each	Locate and defend substantive personal work.

Criterion points = weight × level / 4. Each student receives the shared 80-point score + their own 20.

Home message: A strong project can receive an excellent grade without beating every state-of-the-art model.

How evidence becomes points and a grade

LEVEL ANCHORS KEEP DECISIONS CONSISTENT · USE UNROUNDED TOTALS AT THE BOUNDARIES

THE SAME 0–4 ANCHORS FOR EVERY CRITERION

0	**Missing / invalid**	No usable evidence, or fundamentally invalid evidence.
1	**Weak**	Partial evidence with major gaps or misunderstandings.
2	**Adequate**	Correct, usable evidence that meets the criterion.
3	**Strong**	Well-justified evidence with only minor gaps.
4	**Excellent**	Unusually rigorous or insightful work within the agreed scope.

PROJECT GRADE

0–50 points	2.0
>50–60 points	3.0
>60–70 points	3.5
>70–80 points	4.0
>80–90 points	4.5
>90–100 points	5.0

These are project grades. The current syllabus governs the overall course grade and retakes.

Home message: The score is transparent; incomplete or invalid work needs correction before it can pass.

Mini-Conference: present AND submit the project

CLASS 7 · 5–6 MIN/PERSON INCLUDING SHORT QUESTIONS · 10–12 MIN FOR A PAIR

YOUR FINAL TALK

Question
Why it matters; the anchor paper.

Design
Baseline, comparators, protocol.

Evidence
Main comparison + diagnostic.

Conclusion
Meaning, failures and limits.

Ownership
What each person contributed.

Both team members must speak.

FULL DECK + A SHORT FINAL TALK

FULL PRESENTATION, PPTX + PDF
Submit the complete deck developed throughout the semester.

TALK: 5–6 MIN/PERSON
Show the question, key evidence and conclusions on selected slides.

ALL INVESTIGATED ASPECTS
Explain the methods, experiments and final solution.

FIGURES AND INTERPRETATION
Show plots, illustrations, diagnostics, failures and limitations.

SUPPORTING FILES
Send code, saved results and read-paper PDFs. See the checklist.

Home message: Submit the complete presentation and supporting files; explain your own contribution.

Final submission checklist

CLASS 7: ONE ZIP IN TEAMS BEFORE YOUR SCHEDULED FINAL PRESENTATION

SLIDES.pptx + SLIDES.pdf	The complete presentation developed throughout the semester. All studied aspects, experiments, final solution, charts, and conclusions.
CODE/ + README.md	Code, notebooks, configs and dependencies without models and training data. README: data/model versions and exact run commands. Link to the data.
RESULTS/	Raw metrics (CSV/JSON), saved run outputs and final plots. Record methods, settings, seeds and compute per run.
PAPERS/ + LITERATURE.csv	Full PDFs of the papers you read, matched to the reading matrix. At least 3 different papers/person and 6 different papers for a pair.
IN THE PRESENTATION	Contents/slide index and a concise overview of the main findings. Link important claims to detailed slides, code and saved results.
CONTRIBUTIONS_AI.txt	Each student's contribution in the final solution, with file/run pointers. Brief AI-use statement and how you checked AI-assisted work.

ZIP name: CI_2026_GroupXX_Surname1_Surname2.zip (solo: one surname)

Home message: Send the complete semester presentation, code, saved results and PDFs of the papers you read.

The best projects can become research papers

OPTIONAL FOLLOW-UP · MUTUAL AGREEMENT · STUDENTS LEAD THE WORK AND THE AUTHOR LIST

COURSE PROJECT
A credible question and initial evidence.

EXTENDED EXPERIMENT
Stronger controls and a defensible contribution.

MANUSCRIPT
Students lead experiments and writing.

CONFERENCE / JOURNAL
Submit to a suitable venue; revise after review.

HOW I CAN SUPPORT YOU

Promising projects may receive my support if we both agree.

Students remain first/lead authors.

I support methodology and writing as a contributing co-author. Authorship reflects actual contributions.

Previous student publications appear on the following two slides.

Home message: Publication is a supported opportunity; it is not a condition for an excellent course grade.

Student publications with my support

EXAMPLES FROM PREVIOUS YEARS: 2021–2024. CLICK A TITLE FOR ITS SOURCE.

Jakub Kosno, Daniel Bulanda (ICONIP 2023, published 2024)
Explainable Sparse Associative Self-Optimizing Neural Networks for Classification

Daniel Henel, Aleksander Mazur, Marcin Retajczyk (SSCI 2023)
Neural network for musical data mining for phrase boundary detection

IEEE Travel Grant Award for Daniel Henel: USD 1,000

S. Czaplak (IJCNN 2022)
Automatic Optimization of Hyperparameters Using Associative Self-adapting Structures

M. Wójcik, D. Bulanda (ICONIP 2021)
Associative Graphs for Fine-Grained Text Sentiment Analysis

Home message: Your course project can become the starting point for publishable research.

Student publications with my support

EXAMPLES FROM PREVIOUS YEARS: 2018–2020. CLICK A TITLE FOR ITS SOURCE.

E. Ergün (IJCNN 2020)
YOLOv3 Precision Improvement by the Weighted Centers of Confidence Selection

K. Gołdon (ICANN 2019)
Temporal Coding of Neural Stimuli

M. Janowski (ICAISC 2018)
Supervised Neural Network Learning with an Environment Adapted Supervision Based on Motivation Learning Factors

P. Papież (ICAISC 2018)
Motivated Reinforcement Learning Using Self-Developed Knowledge in Autonomous Cognitive Agent

Home message: You can also do it: a focused project can grow into a research publication.

Fair assessment: evidence plus personal contribution

CLEAR MINIMUMS · NO DOUBLE MARKING · THE SAME RULES FOR EVERY TEAM

BEFORE A PROJECT CAN PASS

Complete	Full presentation, supporting files and individual defence.
Read	≥3 assigned papers/person; ≥6 unique papers in a pair.
Evaluate	A valid main comparison with no known leakage.
Understand	At least Adequate (level 2 in the rubric).
Contribute	At least Adequate (level 2 in the rubric).

PRACTICAL AND PROPORTIONAL

- Five shared criterion scores + two personal scores.
- One complete presentation; an index and targeted questions.
- No separate checkpoint marks or repeated long reports.
- No bonus for hardware cost or publication ambitions.
- Attendance and make-up work follow the current syllabus.

Home message: Shared outputs receive one score; each person must demonstrate understanding and contribution.

Before the next class: bring a research pitch

CLASS 2 · A SHORT PITCH PLUS ONE PAGE OF PLANNING · SUBMIT IN TEAMS

01 Choose 1–3 candidate ideas; recommend one.

02 Draft the question, hypothesis and motivation.

03 Find an anchor paper and accessible candidate data.

04 Plan a baseline, relevant comparators and key metrics.

05 State compute/resources, risks and a fallback.

06 Register your team and assign substantive work packages.

PREPARE AND BRING FOR THE 02 CLASS

3–5 pitch slides
A 1-page project plan
2-3 min/person

READ BEFORE THE MEETING

Class 2: ≥1 full paper/person.
Class 3: ≥3 papers/person.

Pitch slots, exact dates, submission time and the new Teams access details will be posted in Teams.

Home message: Come with a question you care about, an anchor paper and a plan you can realistically execute.